

the silicon material is deposited and etched to form nanocrystals measuring 20 nm. Finally, the gate is formed over the crystal array. Data storage is done by injecting and expelling charge into the silicon nanocrystals. Thus, each nanocrystal can be used to store a single bit of data.

The entire process was done on a 200 mm silicon wafer using methods that employed existing chip fabrication tools. "In effect, we have combined self-assembly with semiconductor fabrication," said Black. Further reduction in the size of the nanocrystals is possible by using polymers with lower molecular weight. Talking about its advantages, Guarini said that since there are so many nanocrystals, even if one loses the charge, the device won't be affected much. That's because, "There are so many more nanocrystals keeping the charge, thus allowing the flash cell to keep working."

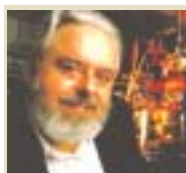
Strained silicon

Originally, work on this technique was done by Judy Hoyt, a physicist at the Massachusetts Institute of Technology. The speed of the flow of current in a chip depends on the structure of the silicon lattice. Each silicon atom has an energy state around it called an orbital. These orbitals merge to allow the current to flow through the silicon lattice. There are six orbitals around a silicon atom; two are in the direction of the current flow while four are perpendicular to it. In normal circumstances, all the orbitals have equal energy so there is no preferred direction when current flows. However, when the lattice is stretched, energy in the two horizontal orbitals decreases, which allows the electrons to flow faster in that direction. Similarly, squeezing the lattice allows positive charges (holes) to flow faster through it.

There are two types of silicon regions in a transistor: n-type and p-type. Silicon, when doped with elements such as phosphorous, acquires extra electrons—this is the n-type silicon. Similarly, silicon doped with elements such as boron acquires a 'hole', which is an electron deficiency—this is the p-type silicon. Therefore, to increase current flow, the n-type areas have to be stretched and the p-type areas have to be compressed. Depositing silicon nitride over the transistor at high temperatures stretches the

silicon lattice. While cooling silicon nitride, it forces the silicon lattice below it to spread out; this stretching helps to speed up the electrons flow by 10 per cent. The p-type region is compressed by carving trenches along its opposite ends and filling them with silicon germanium. As a result, the hole conduction improves by 25 per cent.

The flow of either electrons or holes is called current flow. So, the faster the flow of current, the quicker is the switching of the transistors between the On/Off states, which results in faster processors. All this is achieved without having to reduce the size of the transistors. Implementing the



Dr. Phaedon Avouris
Manager,
Nanometer Scale
Science, IBM
Research Centre

// The basic science is still not totally understood. Nature does use self-assembly, but nature had a research and development time of over two billion years //

strained silicon technique results in an increase in manufacturing costs by just two per cent. Intel has successfully implemented this technique and will use strained silicon in its upcoming processor—Prescott—that will have components measuring 90 nm.

Carbon nanotubes

Carbon nanotubes are extremely thin and very long cylinders of carbon atoms. The typical radius of a carbon nanotube is around a few nanometers; at the same time, their length can be thousand times that of the diameter. Compare this to the components in the Intel Prescott and you get a clear, defined picture of what it is. This nanostructure was discovered by Sumio Iijima, an NEC researcher in the year 1991. Ever since its discovery, there has been a lot of interest in them and a lot of R&D has been going on around this material. This is because carbon nanotubes have an interesting structure that allows them to have some extraordinary properties—they are extremely thin, but stronger than steel. They have potential applications in not just electronics, but in fields ranging from automobiles to constructions. Samsung has developed display prototypes using carbon nanotubes.

Indeed, these nanotubes look like 'one-size-fits-all-solution'.

Pure carbon is found as graphite and diamond. Owing to its atomic structure, diamond is the strongest material. Graphite's layered structure causes it to have both metallic and non-metallic properties, making it more of a semiconductor. The layers, formed by carbon atoms bonded in a hexagonal pattern, are loosely stacked over one another, analogous to a ream of paper. Carbon nanotubes look similar to a sheet of graphite rolled into a cylinder. Due to these structural similarities, carbon nanotubes share the semiconductor properties of graphite.

Like diamond, the bonding of atoms in a carbon nanotube is very strong; thus giving it very good stability. Carbon nanotubes don't suffer from current leakages and thermal problems that besiege the current silicon-based processors. There are two types of nanotubes: Single Wall Nanotube (SWNT) and Multi Wall Nanotube (MWNT). SWNT are structured like normal cylinders, while MWNT have a structure similar to concentric cylinders enclosed within one another. Of these, the former is suitable for usage in the semiconductor industry.

Depending on their molecular structure, carbon nanotubes can be made to act as a metal, insulator or a semiconductor. They should be manufactured in huge quantities with the desired molecular structure. Many techniques have been developed for manufacturing carbon nanotubes, but these are still confined to research labs. Nevertheless, a lot of progress is being made towards making this an affordable alternative to the current technology.

A lot of companies are working on carbon nanotubes: in May 2002, IBM researchers announced that they have created transistors based on carbon nanotubes that perform better than silicon transistors. Nantero, a Massachusetts-based start-up company is developing a nanotubes-based memory. The Nantero technique involves embedding horizontal strands of carbon nanotubes into silicon. Perpendicular to this, vertical strands are suspended. When a suspended carbon nanotube doesn't touch the one at the bottom, electrical resistance is high, which is interpreted as 'zero'. However, when a small electrical charge is applied,